

POORNAPRAJNA COLLEGE (AUTONOMOUS), UDUPI



**A Mini-Project Report
submitted in partial fulfilment
of the
requirements for the award of
the Degree of**



**BACHELOR OF BUSINESS ADMINISTRATION (BBA)
under
MANGALORE UNIVERSITY**

**“SUSTAINABLE SAAS BUSINESS STRATEGY:
INTEGRATING PREDICTIVE MODELLING AND UNIT
ECONOMICS OPTIMIZATION”**

**SUBMITTED TO
MRS. PRATHIBHA S BHAT
Assistant Professor
Department of Commerce and Management**

**NAME OF THE STUDENT
CHIRANJEEVI PK**

**UUCMS REGISTER NUMBER
U05PL23M0032**

DECLARATION

I hereby declare that the project report entitled **“SUSTAINABLE SAAS BUSINESS STRATEGY: INTEGRATING PREDICTIVE MODELING AND UNIT ECONOMICS OPTIMIZATION”** is a record of the original work carried out by me during the academic year 2025-26 under the guidance and supervision of, **Mrs. Prathibha S Bhat, Assistant Professor, Poornaprajna College (Autonomous), Udupi**, in partial fulfilment for the completion of **SIXTH SEMESTER (BBA) of MANGALORE UNIVERSITY**.

I further declare that all the information and data presented in this project have been obtained and presented in accordance with academic rules and ethical conduct. All sources of information, data, and literature used in this project have been duly acknowledged.

Place: Udupi

Date:



Chiranjeevi PK
U05PL23M0032

Poornaprajna College
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This is to certify that **Mr. Chiranjeevi PK**, student of **Final Year BBA**, with Register Number **U05PL23M0032** have successfully completed the project work titled

“SUSTAINABLE SAAS BUSINESS STRATEGY: INTEGRATING PREDICTIVE MODELING AND UNIT ECONOMICS OPTIMIZATION”,

in partial fulfilment of the requirements for the award of the **Degree of Bachelor of Business Administration (BBA)** under **Mangalore University**.

The project report is submitted to the **Department of Commerce and Management**, Poornaprajna College, Udupi, during the academic year **2025-26**.

Further, this is to certify that the above work is a record of the Bonafide work carried out by the candidate under my guidance and supervision. The contents of this report are original and have not been submitted to any other Institution/University for the award of any degree, diploma, or certificate.

Mrs. Prathibha S Bhat
(Project Guide)

Prof. Shivakumar PT
(HOD, Dept of Commerce & Management)

Dr. Ramu L
(Principal)

Place: Udupi

Date:

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Chapter 1 – Introduction

1.1 Overview of SaaS Business Model

The Software as a Service (SaaS) business model is described as “a cloud-based distribution and subscription revenue business model in which software is delivered to the customer in a recurring and remote manner rather than through licenses or on-premise installations.”

In other words, in SaaS, “customers access software applications provided by SaaS companies over the internet rather than through local installations or licenses.” This allows businesses to deliver software to customers at scale without the cost and complexity associated with software infrastructure.

SaaS is considered to be part of the subscription economy business model, which has gained popularity in different industries due to its ability to “shift business strategies to focus less on customer ownership and more on customer relationships.” In SaaS, “every customer represents a stream of revenue over time through subscription fees.” This means that SaaS businesses generate cash flows through subscription fees paid by retained customers. Moreover, SaaS allows businesses to “shift software costs to the customer balance sheets as an operating expense rather than capital expenditures.” This allows SaaS businesses to “reduce barriers to entry for new customers.”

The financial health of SaaS businesses is analysed through a set of financial metrics unique to SaaS businesses. For instance, Monthly Recurring Revenue (MRR) and Annual Recurring Revenue (ARR) are used to measure the recurring subscription revenue streams. On the other hand, Customer Acquisition Cost (CAC) measures the cost to acquire a customer in terms of sales and marketing expenses. Customer Lifetime Value (LTV) measures the value of each customer to the business over time.

One of the biggest challenges for SaaS companies to tackle is the balance between growth and profitability. A high upfront cost for acquiring new customers can lead to rapid growth in terms of the number of users, but poor retention and high CAC in comparison to LTV can put pressure on the financial viability of the business. Effective unit economics, such as maintaining an LTV to CAC ratio of 3 or more, have been recognized in industry analysis as a benchmark for SaaS company sustainability.

Another important metric for SaaS companies to track is the churn rate, which represents the number of customers who have cancelled their subscription to the product within a certain period of time. High churn rates have been recognized in industry analysis as one of the key areas of focus for SaaS business models, and retention has been recognized as a key aspect of SaaS business models. Additionally, the CAC payback period, or the period of time required to pay back the CAC, represents a key metric for SaaS companies, and its importance has been recognized in industry analysis.

Academic and industry analysis of SaaS companies in their early stages have recognized pricing models for SaaS companies to have a significant impact on their financial results. SaaS companies have been able to adopt various pricing models, including tiered, freemium, and usage-based pricing models, and their implications on financial results have been recognized in industry analysis.

1.2 Subscription Revenue Mechanism

The subscription revenue mechanism is the core of the Software as a Service business model, differentiating it from other conventional software products. The SaaS business model does not rely on generating revenues through one-time sales; instead, it generates revenues through subscription-based services, which are paid either monthly or annually. The subscription-based business model has been found to be beneficial in generating revenues, as it helps in long-term financial planning, which is essential in the highly competitive business environment.

Recurring Revenue as the Core Financial Engine

The subscription-based business model generates revenues primarily in the form of Monthly Recurring Revenue (MRR) and Annual Recurring Revenue (ARR). MRR normalizes the revenues generated by the business on a monthly basis, whereas ARR scales the MRR on an annual basis. These measures of recurring revenues have been found to be the most widely accepted metrics in the SaaS industry to track the growth of the business, as well as the stability of the revenues generated by the business (Corporate Finance Institute, 2025). The SaaS business model relies on subscription-based services, thus providing a better estimate of the financial health of the business in comparison to conventional business models that rely on one-time sales to generate revenues.

The SaaS business model has been found to be beneficial in generating revenues in a stable manner by reducing the fluctuations in revenues that are common in conventional business models. According to industry analysis, companies that exhibit higher MRR growth rates have been found to be better positioned to attract investments in the business, as the long-term financial health of the business can be accurately determined.

Subscription Fee Structures

The SaaS business model has been found to implement different subscription fee structures depending on the target segment of the business, the value provided to the customers, and the pricing strategy of the business. The different subscription pricing strategies that the SaaS business model has been found to implement in the industry are as follows:

Tiered Pricing: The tiered pricing strategy has been found to be the most widely accepted pricing strategy by the SaaS industry, as it allows differentiation of the product features according to the needs of the customers.

Freemium Model: The basic features are provided for free to encourage more people to use it, and then money is made by upgrading to premium features. The freemium model has been successful in increasing user base at a rapid pace, but it should be monitored to see how many users are upgrading to premium features.

Usage-Based Pricing: The price is charged according to how much service has been used by the consumer. This type of pricing is gaining popularity in SaaS segments like cloud infrastructure and API-based products.

Revenue Recognition in SaaS

Revenue recognition in SaaS accounting has certain principles to be followed. The main principle in SaaS accounting is to recognize revenues over the time period in which the service has been delivered

rather than at what time it was sold to the consumer. This type of accounting has been approved by international financial reporting standards and gives a realistic view of business operations.

Subscription Renewal and Its Impact

Another important part of subscription-based revenue generation is renewal rate, which measures how many subscribers have renewed their subscription at the end of each subscription period. The renewal rate has a direct effect on churn rate, which measures how many subscribers have cancelled their subscription.

SaaS companies utilize analytics as well as customer success strategies to boost renewal rates. This is an example of the intertwining of the subscription revenue model with other aspects such as predictive analytics as well as the analysis of customer behaviour.

Customer Lifetime Value (LTV) and Contribution Margin

The two metrics that can be derived from the subscription revenue model as well as relate to the aspect of revenue are the Customer Lifetime Value (LTV) as well as the contribution margin. The LTV is the estimate of the revenue that can be earned from a customer over the entire subscription duration. The contribution margin, on the other hand, is the estimate of the revenue that can be earned after the costs of the subscription have been covered. These metrics form the core of the determination of whether the subscription revenue model can be utilized to grow the business in a sustainable manner.

Benefits as well as Challenges

The subscription revenue model has a number of benefits, including the aspect of predictable cash flows, improved customer retention strategies, as well as higher valuation multiples, as companies that utilize the subscription revenue model attract higher valuations from investors. However, the subscription revenue model also has a number of challenges that can be attributed to it.

1.3 Growth vs Profitability Dilemma in SaaS

The SaaS industry has the characteristic of rapid revenue growth potential, driven by the scalability of the cloud-based delivery model of software products. However, the growth of the business also implies that profitability may not be achieved in the same proportion, thus posing a challenge to the sustainability of the business in the long run. It has been a common observation among researchers that the focus on revenue growth may not always be in line with the actual profitability of the business, as the cost of acquiring new customers may not be matched by the value of the business that is returned by the customer in the long run (SaaS unit economics and churn studies).

The LTV to CAC ratio is one of the primary measures that determine the sustainability of the business in the SaaS industry, as it compares the Customer Lifetime Value (LTV) with the Customer Acquisition Cost (CAC) of the business. The commonly accepted norm in the industry is that the LTV to CAC ratio of a SaaS business should be around 3:1, i.e., the lifetime value of the customer should be three times the cost of acquiring the customer (SaaS unit economics and churn studies).

High growth rates of the business may not always be a true indicator of the financial health of the business, as the churn rate may also be high, thus affecting the profitability of the business in the long run. The churn rate is the rate at which the customers of the business stop using the product, thus requiring the business to spend more on acquiring new customers to replace the ones that have churned out of the business. The churn rate has a significant impact on the profitability of the business, as the LTV of the business may not be enough to cover the cost of acquiring new customers.

Another metric commonly used to balance growth and profitability is the Rule of 40, which holds that the sum of growth rate and profitability should be above 40%. The metric has been widely accepted in academic and professional circles to determine whether growth-oriented investments are viable in comparison to profitability gains. Enterprises with high growth rates and low profitability may still be considered viable if the combined metric meets or surpasses 40%, while those with low growth and profitability may be in trouble.

Another metric used to determine how SaaS businesses balance growth and profitability is Net Revenue Retention (NRR), which has gained prominence in recent times as a key metric to determine how SaaS businesses balance growth and profitability. A business with NRR above 100 percent indicates that growth in revenues from existing customers through upselling or cross-selling is above revenues lost due to churn or downgrading.

Another academic study indicates that investment strategies and priorities vary depending on the maturity level of SaaS businesses. For instance, SaaS businesses in the early stages of growth focus primarily on customer acquisition and revenue acceleration, even if it means sustaining losses in terms of profitability to gain significant traction in the market. However, at later stages, SaaS businesses need to be profitable to sustain operations and seek funding or go public.

1.4 Problem Statement

Despite the high growth rate and potential revenue growth of Software as a Service business models, it has been observed that many SaaS businesses face challenges related to financially sustainable growth, as revenue growth can be a result of inefficient unit economics and retention strategies. While conventional software businesses can be sustained on a single transactional basis, SaaS businesses require high levels of recurring revenue and engagement, and hence metrics such as churn rates, Customer Acquisition Cost (CAC), Customer Lifetime Value (CLV), and revenue retention are critical to the viability of SaaS businesses. High rates of churn and increasing costs of acquisition can impact the lifetime value of customers and create a situation of financial stress, even if revenue growth is high. This creates a strategic dilemma for SaaS businesses, as they need to strike a fine balance between growth and unit economics.

Existing research has demonstrated how inefficient retention strategies and high rates of churn can impact revenue and profitability, and create a situation wherein businesses have to spend more on growth to maintain growth rates that are higher than financial returns. This situation is a result of strategic decisions made on growth, without considering economic viability and unit economics. Moreover, the absence of predictive tools and techniques to understand customer behaviour and market conditions creates a situation wherein conventional metrics and tools may not be able to forecast revenue and financial growth, and this situation can create a challenge for strategic decision-making.

1.5 Research Objectives

- To assess the efficiency of various predictive modelling techniques in predicting customer churn for SaaS products using historical subscription-based data.
- To understand the relationship between key unit economic metrics for a business, such as Customer Acquisition Cost (CAC), Customer Lifetime Value (LTV), and churn rates, in relation to long-term sustainability for a SaaS business.

- To understand the relationship between predicted churn probability and business financials, thereby enhancing decision-making for long-term sustainability.
- To generate actionable business recommendations for a SaaS business based on a combined analysis of both predictive modelling for customer churn and unit economic evaluation.

Chapter 2 – Importance and Scope of the Study

2.1 Importance of Predictive Analytics in SaaS

Predictive analytics has been recognized as one of the essential capabilities for Software as a Service (SaaS) companies, considering the high volume of data available on customer behaviour, subscription, and usage in cloud-based services. Predictive analytics can be defined as the process of predicting possible results in the future using various statistical models, machine learning algorithms, and historical data patterns. Predictive analytics helps SaaS companies transform their historical data into useful insights to make informed business decisions.

The most prominent application of predictive analytics in SaaS companies involves predicting customer churn. In SaaS companies, retaining customers has a direct impact on the stability of the company's revenues and its profitability in the long term. According to various studies, predictive analytics has been recognized for its potential to identify potential customers who might churn in the near future by analysing their behavioural patterns and usage of services. Predictive models help companies to apply retention strategies for their potential churners, resulting in higher revenues and customer lifetime value.

Predictive analytics helps SaaS companies in predicting their revenues and subscription trends. SaaS companies operate on subscription-based models, and their revenues depend on subscription models. Therefore, predictive models help companies in predicting their revenues and subscription trends in the future. Predictive models can analyse historical subscription and churn data to predict possible revenues and their growth in the future. According to various studies on predictive analytics in SaaS companies, predictive models have been recognized for their potential to increase the accuracy of revenues forecasting models by including churn prediction variables in predictive models.

Another significant advantage of using predictive analytics in SaaS is that it can help the organization develop an appropriate strategy for the segmentation of customers. In this context, it can be mentioned that the advanced predictive models can analyse the data related to the usage of the SaaS product. By using the advanced predictive models, the organization can identify the segments of customers that can potentially increase the revenue for the organization. In this context, it can be mentioned that recent studies have mentioned that the use of predictive segmentation can help the organization increase the average revenue generated from each user.

In this context, it can be mentioned that the recent advancements in artificial intelligence have helped enhance the role of predictive analytics in the SaaS ecosystem. In recent studies have mentioned that the use of artificial intelligence can help the organization develop an appropriate predictive model that can analyse large amounts of data. By using the advanced predictive model, the organization can enhance the forecasting of the demand for the SaaS product.

2.2 Importance of Unit Economics in Financial Sustainability

The role of unit economics in the financial sustainability of SaaS business organizations is significant. SaaS business organizations do not rely on the conventional business model of selling software products to customers, as in the case of other software companies. Instead, SaaS business organizations offer subscription-based services to customers, implying that the organization has to consider the profitability of every customer it acquires in the long run. The application of unit economics helps business organizations analyse whether the revenue earned from a customer is higher than the cost of acquiring the customer, thus providing a measure of sustainability of the business organization.

The application of unit economics in SaaS business organizations also involves the relationship that exists between the Customer Acquisition Cost (CAC) and the Customer Lifetime Value (LTV) of the business organization. The CAC of the business organization refers to the cost of acquiring a new customer, including the cost of the sales force, the cost of the marketing team, as well as the cost of promotions. The LTV of the business organization refers to the revenue that the organization expects to earn from the customer over the entire duration of the subscription period. According to research on SaaS business models and revenue forecasting, the sustainability of the business organization relies on the relationship that exists between the LTV and CAC of the business organization because it ensures that the business organization is not growing in terms of revenue alone but also in terms of profitability.

The application of unit economics in SaaS business organizations also involves the churn rate of the business organization, which refers to the number of customers that the business organization loses over a specified duration of time. The application of unit economics in SaaS business organizations implies that the churn rate of the business organization has a significant impact on the profitability of the business organization in the long run, as studies on SaaS subscription business models reveal that the churn rate of the business organization has a significant impact on the predictability of the revenue earned by the business organization.

Unit economics may also be used in making strategic decisions regarding price strategies, marketing strategies, and customer retention strategies. For instance, SaaS firms may invest heavily in acquiring customers during the growth stage, expecting that in the long term, subscription fees may offset these acquisition costs. However, without unit economics analysis, growth strategies may lead to increasing operational costs without any corresponding profitability. As such, businesses need to regularly check whether each customer acquired adds value to the business after deducting acquisition costs and service delivery costs.

Therefore, unit economics may be used as an analytical tool in making strategic decisions regarding financial sustainability in SaaS businesses. For instance, SaaS firms may use unit economics to determine whether growth strategies are creating long-term value in the business through financial indicators like CAC, LTV, churn rate, and contribution margin. The integration of financial indicators with predictive analytics may be used to make strategic decisions regarding financial sustainability in SaaS businesses through predictions regarding future revenues and customer retention strategies.

2.3 Industry Relevance

The relevance of Software as a Service business models in the global technology sector can be understood in terms of their increased acceptance in the global technology sector as a result of

increased business use of software delivery in a cloud-based environment. Software as a Service is a software delivery model in which businesses can use software applications through internet-based subscription services. Software as a Service business models have been recognized in various business studies related to software delivery systems in terms of their ability to allow businesses to use software applications in a subscription-based environment.

The increased growth of the Software as a Service sector is a clear indicator of its relevance in terms of its acceptance in the global digital economy. Market research indicates that the Software as a Service sector was valued at approximately USD 315.68 billion in 2025 and is projected to increase to nearly USD 1.13 trillion in 2032. Software as a Service business models have been recognized in terms of their ability to provide software delivery systems in a subscription-based environment for businesses.

Some of the key technology companies that have been recognized for their ability to provide Software as a Service business models include Salesforce, Adobe Systems, and Microsoft. Software as a Service business models have been recognized in terms of their ability to provide software delivery systems for businesses in a subscription-based environment. For instance, Microsoft is recognized as a leader in terms of overall revenue in the enterprise software as a service sector.

Another aspect that can be seen as a criterion for the relevance of the industry is the rapid growth of the subscription economy, which encompasses SaaS platforms, streaming services, and other digital platforms that offer subscription-based business models. Studies have shown that subscription-based business models have grown much faster than other industries, thus showing the rising demand for flexible service-based software solutions. The growth of the subscription economy has also led to companies investing heavily in analytics, AI, and predictive technologies to enhance customer retention strategies as well as optimize revenue generation strategies.

Another aspect that has been seen in the industry is the rapid growth of the SaaS industry, thus providing companies with the opportunity to offer SaaS platforms to organizations that rely on digital technologies in different industries such as finance, health, education, and e-commerce services. The SaaS industry has been seen as a critical aspect of modern digital business as it has been preferred by many organizations due to the reduction of costs, rapid application deployment, as well as the ability to offer services to different devices in the organization. The rapid growth of the SaaS industry has led to the need to understand the sustainability of SaaS companies, thus providing a significant research question regarding the integration of predictive analytics as well as the evaluation of unit economics to offer insights into the strategies that SaaS companies can adopt to enhance sustainability in the industry.

2.4 Scope and Limitations

The scope of this study is centred on an examination of how predictive modelling and unit economics can be used to inform sustainable growth strategies for Software as a Service enterprise. The scope of this research is mainly concerned with an examination of critical financial and operational metrics that can impact the sustainability of subscription-based business models.

Some of these metrics include Customer Acquisition Cost, Customer Lifetime Value, and recurring revenue models such as Monthly Recurring Revenue.

Moreover, this study has also explored how predictive models can be used to forecast customer behaviour, such as churn probability and revenue growth, by analysing data obtained from Software as a Service enterprise. By incorporating unit economics into predictive modelling, this study has

demonstrated how data-driven decision-making can be used to inform strategic planning for Software as a Service enterprise.

Research Boundaries

Although the SaaS sector comprises a wide range of products and services in various fields, the current research is focused on customer behaviour analysis and financial performance indicators in subscription-based SaaS products. However, the current research does not intend to evaluate other types of cloud computing infrastructure or platform-as-a-service models, as their operational dynamics differ from subscription-based models.

Moreover, the current research is focused on quantitative analysis of subscription-based metrics and does not cover the qualitative evaluation of organizational strategies, management practices, and competitor models. The purpose of the current research is to evaluate how predictive modelling and unit economics metrics interact in subscription-based SaaS models and affect business growth.

Dataset Limitations

The current analysis is based on a structured SaaS model dataset containing subscription-based metrics, customer behaviour indicators, and financial metrics. Such a dataset is a valuable tool for understanding various customer behaviour patterns. However, a structured dataset may not cover all factors that affect subscription-based decisions in a business environment.

The second limitation is related to the availability of a sufficient amount of data for analysis. In most cases, publicly available or simulated datasets may not represent all factors related to subscription-based models in enterprise-level SaaS products, where millions of customers generate diverse behaviour patterns. Hence, readers should understand that the current research is focused on providing analytical predictions and not universally applicable predictions for all types of SaaS models.

Methodological Scope

The research methodology is centred on various techniques for analysing quantitative data. Such techniques include descriptive statistics, correlation analysis, predictive modelling, and unit economics evaluation. Additionally, predictive models are used for estimating customer churn probability and revenue trend forecasting using subscription-based data.

Although machine learning techniques can offer much deeper and stronger predictive capabilities for business applications, the methodological scope of this research is limited to interpretable models for supporting business analysis.

Chapter 3 – Literature Review

3.1 Studies on SaaS Growth Models

The rapid adoption of the Software as a Service (SaaS) model has gained considerable academic attention, particularly with regard to the growth models of the SaaS business. Various studies have explored the growth models of the SaaS business from different perspectives, including pricing, market adoption, scalability, and financial viability. The SaaS model, in general, is considered one of the most important changes in the software industry, as the SaaS business model changes the focus of the software business from one-time license revenue to recurring revenue.

One of the most important areas of academic research on the growth models of the SaaS business is the economic and operational characteristics of the SaaS business. Research published in the operations management and information systems literature indicates that the growth of the SaaS business is closely related to recurring revenue models. Research published in the journal of Production and Operations Management indicates that the growth models of the SaaS business are determined based on factors such as pricing, competition between the software business, and customer adoption, which have an impact on the growth of the SaaS business

Another important aspect of the academic research on the growth models of the SaaS business is the strategic advantage that the SaaS business model provides to the organization. Research on the SaaS business model indicates that the SaaS business model provides strategic advantage to the organization, particularly with regard to the efficiency of the business. Research on the adoption of the SaaS business model in business organizations indicates that the cloud-based software service reduces the infrastructure cost of the organization. As such, the SaaS business model provides strategic advantage to the organization, particularly with regard to the growth of the business.

Moreover, the recent literature also highlights the significance of data-driven decision support and analytics in the context of SaaS growth strategies. As discussed in the literature, the SaaS model generates a huge amount of user data, which is used to analyse the user behaviour and system performance. Researchers have proved that the data-driven decision support system is essential in the context of SaaS growth strategies, as it would help the organization to retain the customer and sustain the growth in the subscription-based software industry.

Another important aspect of the SaaS growth model is the pricing and monetization strategies adopted by the organization. Researchers have used the academic studies on the pricing model and proved that the SaaS model has adopted a flexible pricing strategy, which includes tier pricing, usage-based pricing, and dynamic pricing. Using this pricing model, the organization could efficiently generate the required revenue and increase the level of competitiveness in the industry. Recent studies on the pricing model also highlight the adoption of artificial intelligence-based pricing strategies, which would help the organization to optimize the subscription-based pricing model.

Moreover, the studies on the SaaS model also highlight the overall impact of the SaaS model on the overall economy of the organization. Researchers have proved that the overall adoption of the SaaS model would help the organization to increase the level of operational flexibility and support the digital transformation strategy. As the overall system is based on the cloud model, the organization could easily change the operations based on the changing demands of the industry, and the overall adoption of the SaaS model would act as a catalyst in the context of the overall growth of the organization in the industry, especially in the finance, healthcare, and technology sectors.

3.2 Research on Customer Churn Prediction

Customer churn prediction is a significant research area in subscription-based industries like Software as a Service (SaaS), telecommunication services, etc. Customer churn is a phrase used to describe a state where a customer terminates the subscription or stops the business relationship with a particular service provider or company. Software as a Service (SaaS) is a subscription-based business model, and these businesses depend on a revenue model to ensure sustainability, so prediction of customer churn is essential to ensure the sustainability of such businesses. Researchers emphasize customer churn prediction as a vital tool for ensuring the sustainability of such subscription-based services, as it

helps companies identify customers who are likely to terminate their subscription, thereby ensuring timely intervention to retain customers.

Recent research indicates the increasing application of machine learning and predictive analytics in customer churn prediction models for subscription-based services such as SaaS platforms. Customer interaction is a major source of data in subscription-based services such as Software as a Service (SaaS) platform, and such data is generated as customers interact with the product or services offered by the company. Analysing such data using predictive analytics helps companies understand customer behaviour patterns associated with customer churn prediction. Machine learning techniques such as logistic regression, decision tree, and ensemble techniques have been applied in customer churn prediction, and such techniques help companies classify customers as those who will churn and those who will remain with the company, thereby anticipating customer attrition and ensuring timely intervention to retain customers.

Research focused on customer churn prediction in Software as a Service (SaaS) platform indicates the application of customer churn prediction models is essential for improving business results, as such models help companies proactively engage with customers to ensure customer retention. Studies focused on customer predictive modelling for Software as Service (SaaS) products indicate that such models are essential for improving business results, as they help companies understand customer behaviour patterns and predict early warning signals of customer churn prediction, thereby ensuring timely intervention to retain customers who are likely to terminate their subscription with the company.

The other significant aspect of the research in churn prediction is the identification of the major factors that affect customer churn. The studies based on the application of the research have indicated that the major factors affecting customer churn include customer tenure, usage intensity of the services, pricing of the subscription, and customer support interactions. The incorporation of these factors in the model has indicated the potential of the application of the model in predicting customer churn with high precision.

The recent research studies have also indicated the application of optimization techniques in the prediction of customer churn in the SaaS model. The studies based on the application of optimization algorithms in the prediction model have indicated the potential of the application of feature selection techniques in the prediction of customer churn in the SaaS model, leading to the increased significance of the application of the combination of the model with the data preprocessing techniques.

3.3 Revenue Forecasting Methods

Revenue forecasting is a vital part of the financial planning and strategic planning process in the Software as a Service industry. Software as a Service is different from other software organizations, as most software organizations operate on a one-time sale model, whereas Software as a Service operates on a subscription-based model, in which future revenue is heavily dependent on customer retention and subscription renewals. Hence, it is critical to forecast the revenue in the Software as a Service industry in order to assess the sustainability of the business.

Research on the Software as a Service model has revealed that the process of forecasting the revenue is critical and is also related to the subscription lifecycle management process. In a subscription-based model, the subscription revenue is impacted by several factors, and it is critical to forecast the

subscription revenue based on the customer retention and subscription renewals. Hence, it is critical to forecast the subscription revenue, and the forecasting model has to incorporate the customer retention and subscription model in order to forecast the revenue. (researchgate.net)

One of the most common methods of forecasting the revenue in the Software as a Service industry is the time series method, in which the historical trend of the revenue is used in order to forecast the future revenue. In the time series method, the historical subscription-based model is used in order to forecast the future revenue, and this method is critical in the Software as a Service industry, as the subscription-based model generates a lot of financial data in the form of Monthly Recurring Revenue and Annual Recurring Revenue.

Another key forecasting method is the use of predictive analytics and machine learning techniques. This method of forecasting has the capability to analyse complex customer behavioural patterns and determine the variables affecting the fluctuations in revenue generation. This would allow the organization to make more precise predictions about the revenue generation in the near future by using the customer behavioural parameters in the forecasting model.

Studies have also suggested that the use of churn prediction variables in the revenue forecasting model improves the overall predictability of the forecasting model. Churn directly affects the recurring revenue generation of the organization. Therefore, the forecasting model that uses the churn prediction variables would allow the organization to make more precise predictions about the revenue generation in the near future.

In addition, the revenue forecasting model for the SaaS business uses unit economics parameters like Customer Acquisition Cost (CAC) and Customer Lifetime Value (LTV). These parameters would allow the organization to determine the economic viability of the revenue generation in the near future. Therefore, the use of the revenue forecasting model in combination with the unit economics parameters would allow the organization to make more precise predictions about the revenue generation in the near future.

3.4 Studies on LTV, CAC, Contribution Margin

Some of the essential financial parameters that need to be considered for ensuring the sustainability of the Software as a Service model include Customer Lifetime Value (LTV), Customer Acquisition Cost (CAC), and the contribution margin. The combination of the above three parameters is essential for ensuring the viability of the unit economics, which is vital for determining the viability of the customer acquisition and retention strategies for ensuring profitable results for the organization. The relationship between Customer Lifetime Value, Customer Acquisition Cost, and the contribution margin is essential for ensuring the sustainability of the Software as a Service model for the organization.

The Customer Lifetime Value is the amount of revenue that is expected to be generated from the customer during the subscription period with the organization. The studies conducted on the financial parameters of the Software as a Service model revealed that customer retention is a vital factor for ensuring customer lifetime value, which is essential for ensuring profitable results for the organization. The customer subscription history is essential for determining the customer lifetime value, which is vital for ensuring profitable results for the Software as a Service model.

The Customer Acquisition Cost is the financial parameter that is vital for determining the overall expenses incurred by the company for acquiring new customers. The Customer Acquisition Cost includes the overall expenses incurred for marketing, advertising, and promotional activities for acquiring new customers for the organization. Studies on revenue forecasting for Software as a Service models reveal that the relationship between customer lifetime value and customer acquisition cost is vital for ensuring profitable results for subscription-based services. If the customer acquisition costs for Software as a Service models are higher than the customer lifetime value, the organization may experience revenue growth, but it may not result in profitable results for the organization.

The other important metric under unit economics for SaaS is contribution margin, which is the percentage of revenue available after deducting the variable costs of providing the service. Contribution margin helps to understand how much of the revenue generated per subscription is contributing to fixed costs and profits. For a SaaS business, contribution margin is an important metric, as revenues need to be sufficient to cover costs for a long time. Studies show that companies with high contribution margins are more likely to attain sustainable growth and reinvest in product innovation and customer experience improvement.

The relationship between LTV, CAC, and contribution margin is an important tool for understanding the financial efficiency of SaaS companies. Studies by academic researchers show that for a SaaS company, achieving sustainability in growth is possible only by ensuring the LTV and CAC ratio is appropriate, i.e., the revenue generated per customer is significantly higher than the cost of acquiring the customer. By using all these metrics, companies can understand whether their subscription-based business is financially viable or not.

Moreover, the integration of unit economics with predictive analytics is also an important tool for decision-making for SaaS companies. By using predictive analytics, companies can predict customer churn and estimate customer lifetime value, and this is an important tool for achieving sustainability in growth for SaaS companies.

3.5 Identified Research Gap

Based on the review of the existing literature related to the SaaS growth model, churn prediction, revenue forecasting, and unit economics, the following gaps in the literature were observed:

Lack of integration of predictive analytics and financial metrics

Most of the literature focused on the development of the churn prediction model and revenue forecasting model separately. However, little emphasis was given to the integration of the predictive analytics model with the unit economics metrics such as Customer Lifetime Value (CLV), Customer Acquisition Cost (CAC), and contribution margin.

Predictive analytics rather than financial sustainability

Most of the literature related to the development of the churn prediction model focused more on the improvement of the accuracy of the machine learning model rather than the financial sustainability of the SaaS company.

Lack of integration of customer behaviour analytics and economic sustainability

Although the literature related to the customer behaviour analytics was well explored in the context of the SaaS environment, little emphasis was given to the integration of the customer behaviour analytics with the economic sustainability of the company.

Need for developing the decision support model for the management of the SaaS company

Although the theoretical literature related to the SaaS business model was well explored, the need for developing the decision support model for the management of the SaaS company still exists.

Chapter 4 – Research Methodology Review

4.1 Quantitative Analytical Study

The research will adopt a quantitative analytical research approach to investigate the relationship between predictive modelling techniques and unit economic indicators in Software as a Service business environments. Quantitative research approaches are appropriate for this research study because Software as a Service business environments generate numerical data related to customer subscription rates, revenue generation metrics, and customer engagement patterns. This numerical data provides a platform for applying various research tools to identify patterns and generate meaningful insights related to the research problem.

In Software as a Service business environment, various performance indicators such as churn rates, customer lifetime value, and customer acquisition rates are normally stored as numerical data. Quantitative research approaches provide a platform for analysing these numerical data to understand their impact on revenue sustainability. By analysing the patterns related to customer subscription rates, the research aims to identify the patterns that contribute to the long-term financial sustainability of Software as a Service business organization.

The quantitative analytical method also lends support to the application of predictive modelling techniques that can be applied to forecast future outcomes based on historical trends. Research on the application of SaaS business models also indicates the need to incorporate the application of predictive modelling techniques to forecast customer churn rates, revenue forecasts, and customer lifetime value. Quantitative research methods on the application of SaaS business models indicate the need to incorporate the application of quantitative forecasting techniques to gain insights into customer subscription life cycles.

Moreover, the application of the quantitative analytical method also allows for the incorporation of financial unit economics with the application of predictive modelling techniques. Based on the analysis of the relationship between customer churn rates, customer acquisition costs, and customer lifetime value, the research aims to evaluate the application of customer acquisition strategies to achieve business sustainability. Statistical analysis techniques applied to the research also allow for the application of a structured analysis of the relationship between the variables applied to the subscription-based business model.

4.2 Data Source

1. Primary Dataset Source

In this research, the primary dataset source is the publicly available IBM dataset titled "Telco Customer Churn" from the Kaggle platform.

This dataset contains customer information from a telecom service provider with about 7,000+ customer records. Each record represents customer demographics, subscription information, billing information, etc. Though the dataset belongs to the telecom industry, it resembles a software as a

service industry due to the similarities between the billing information for telecom services and the subscription information for software services.

Several attributes from the dataset describe the customer information with respect to the subscription service. These attributes include the duration for which the customer has been with the service provider, the billing amount for the customer every month, the revenue earned from the customer, the type of service plan the customer has taken from the service provider, etc. These attributes create a structured view of the customer subscription service that can be analysed with the help of predictive analysis techniques.

2. Justification for Using a Subscription Proxy Dataset

Access to customer information from software as a service industry is extremely restricted due to confidentiality issues. As a result, researchers often rely on a subscription service proxy dataset from the telecom industry due to the similarities between the telecom industry and the software as a service industry.

Several similarities exist between the telecom industry and the software as a service industry:

- recurring subscription revenue model
- customer relationship contracts
- customer lifecycle management
- customer churn risk management

Due to the similarities between the telecom industry and the software as a service industry, telecom industry customer churn datasets are often used for research studies related to customer retention modelling, predictive analysis, etc.

3. SaaS Variable Engineering

In order to ensure that the telecom data set is in line with a SaaS business scenario, a number of data transformations were carried out on the existing data set variables. These transformations ensure that the data set is in line with various SaaS business data elements.

Data transformations carried out:

Monthly Recurring Revenue (MRR)

Data values related to monthly revenues were considered to calculate recurring revenues per customer.

$$MRR_t = \sum_{i=1}^n Monthly_Revenue_i$$

Customer Lifetime Value (LTV)

Customer LTV was calculated using a relationship between monthly revenues and customer tenure.

$$LTV = Monthly_Revenue \times Tenure$$

Customer Acquisition Cost (CAC)

Since this data set does not provide any acquisition costs, a realistic data variable related to acquisition costs was assumed using proportional costs related to marketing costs.

$$CAC_i = k \times Monthly_Revenue_i$$

Contribution Margin (CM)

Contribution margin values were calculated using service delivery costs related to subscription revenues.

$$CM_i = Monthly_Revenue_i - Service_Cost_i$$

LTV to CAC Ratio

The LTV:CAC ratio is commonly used in SaaS analytics to measure the viability of investments in customer acquisition. A high LTV:CAC ratio shows that the revenue earned from the customer is much higher than the cost of acquiring the customer. It is essential for the growth of a subscription-based business model to have a balanced relationship between LTV and CAC. It has been a benchmark in the industry that a successful SaaS business must have an LTV:CAC ratio of at least 3, which means the LTV earned from the customer must be at least three times the cost of acquiring the customer.

$$LTV : CAC = \frac{Customer\ Lifetime\ Value}{Customer\ Acquisition\ Cost}$$

This data transformation ensures that the data set is in line with various SaaS business data elements.

4. Time Dimension for Revenue Forecasting

In order to ensure a data set suitable for a revenue forecasting scenario, a data variable called "billing month" was generated using existing data set values related to "tenure."

This data set transformation ensures a data set suitable for a time series data set related to subscription revenues, thus enabling a data set suitable for a time series data set related to subscription revenues.

This data set transformation ensures a data set suitable for a time series data set related to subscription revenues.

5. Integration with SaaS Industry Context

While the dataset represents telecom subscription information, the analytical framework applied by the research follows the financial and operational metrics applied by the leading SaaS companies such as:

- Salesforce
- Adobe
- HubSpot

These companies are utilizing various key performance indicators to measure the performance of the business, including churn rate, customer lifetime value, customer acquisition cost, and revenue growth rate. Therefore, the study aligns the analytical approach with the actual business evaluation approach through the application of the same key performance indicators to the subscription proxy dataset.

Resulting Dataset Structure

As a result of the preprocessing and feature engineering steps, the resulting analytical dataset includes features from the following areas:

- Customer lifecycle
- Subscription revenue
- Behavioural
- Churn outcomes
- Engineered SaaS financial

4.3 Variables Used

This study makes use of both original variables provided by the dataset and SaaS financial variables that were created in the analysis. All the variables used in the analysis were categorized into independent variables, dependent variables, and derived financial variables.

1. Independent Variables

Independent variables in the analysis include the variables that affect customer behaviour.

Independent variables in the analysis include:

Tenure – This is the number of months that the customer has been subscribed to the service. This variable is used to denote the length of time that the customer has been with the company.

Monthly Charges – This is the amount that is charged by the company each month to the customer. This variable is used to denote the amount that is charged by the company each month.

Contract Type – This is used to denote the type of contract that the customer has. This variable is used to denote the type of contract that the customer has. It can include month-to-month, one-year contract, or two-year contract.

Payment Method – This is used to denote the method used by the customer in making payments. This variable is used to denote the method used by the customer in making payments.

Internet Service Type – This is used to denote the type of service that is provided to the customer. This variable is used to denote the type of service that is provided to the customer.

2. Dependent Variable

The dependent variable is used to denote the variable that is being analysed in the study.

Churn – This is a binary variable that identifies customers who terminated their services during the study period. The prediction of customer churn is one of the main objectives of this research, as it is useful in determining factors that lead to customer churn.

This variable is used as the target variable in the predictive modelling stage of analysis.

3. Derived SaaS Financial Variables

Besides the variables in the original dataset, financial variables that are commonly used in SaaS analytics were derived. The use of these variables is essential in evaluating customer profitability. The variables include:

Monthly Recurring Revenue (MRR) – This is a variable that identifies revenue obtained from customers who have an active subscription at a given time.

Customer Lifetime Value (LTV) – This is a variable that identifies revenue expected to be obtained from a customer throughout their subscription period, calculated using the relationship between monthly revenue obtained from customers and customer tenure.

Customer Acquisition Cost (CAC) – This is a variable that identifies the expected cost of acquiring a customer using different marketing approaches.

Contribution Margin (CM) – This is a variable that identifies revenue obtained after subtracting service delivery costs from subscription revenue.

LTV to CAC Ratio – This is a variable that identifies the effectiveness of customer acquisition approaches by comparing revenue obtained from customers with the expected customer acquisition cost.

4.4 Tools Used

Analytical procedures in this research were carried out using various computational tools that offer libraries for data analysis, quantitative analysis, and predictive modelling.

1. Python Programming Language

The main programming language used in carrying out the analysis in this research is Python. Python is an effective programming language that offers an environment in which various data analysis techniques can be implemented.

Various Python libraries were used in carrying out the analysis in this research. They include:

- **Pandas:** This library was used in carrying out data preprocessing, cleaning, and transforming the dataset. Pandas is an effective library in carrying out data preprocessing.
- **NumPy:** This library was used in carrying out numerical computations that were necessary in the statistical analysis.
- **Scikit-learn:** This library was used in carrying out predictive modelling techniques, as discussed in the research.

2. Google Collab Environment

The analysis in this research was carried out in the Google Collab environment. This is an environment that enables the execution of Python code in an interactive environment. Google Collab is an effective environment that enables combining code, data results, and text in one document. This is an effective environment in carrying out data analysis in that it enables the creation of an interactive notebook document. This document enables the creation of an effective document that describes various data analysis procedures.

3. Data Visualization Libraries

For presenting the findings of the analysis in a manner that is easily interpretable, data visualization techniques were employed. For this purpose, data visualization libraries in Python were used.

Matplotlib: This library was employed for developing charts and plots, which could be used for illustrating revenue patterns, customer distribution, as well as model evaluation.

Seaborn: This library was employed for developing advanced statistical charts, as well as for better visualization of data patterns.

4. Microsoft Excel

For initial data inspection, as well as verification of calculated financial variables, Microsoft Excel was employed. The utility of Microsoft Excel is that it provides a simple platform for inspecting data, verifying variables, as well as cross-checking computations of different financial variables of SaaS businesses, including MRR, LTV, CAC, and contribution margin.

4.5 Analytical Techniques

This study makes use of a set of quantitative analytical techniques to investigate customer behaviour, revenue models, and financial sustainability within a subscription-based business. The analytical techniques used in this study have been carefully chosen to fit the objectives of the study. The stages include:

1. Descriptive Analysis

Descriptive analysis was carried out to gain a basic understanding of the nature of the data set and to identify the various patterns associated with customer behaviour. Descriptive analysis was used to identify the various variables associated with customer behaviour. The variables used in this study include customer behaviour, subscription charges, contract type, and churn distribution. Descriptive analysis was used to identify the relationships between various variables that can be used to identify the various factors that influence customer behaviour.

2. Churn Prediction Model

A churn prediction model was developed to forecast the likelihood of customers leaving the subscription service. Churn prediction is a critical factor for any subscription-based business model as the customer base directly impacts the revenue generation.

Machine learning algorithms were implemented to classify customers into churner and non-churner groups based on their attributes. Factors such as customer tenure, subscription charges, contract types, and payment modes were considered for the model. The predictive model will help to identify the factors that influence customer churn.

3. Revenue Forecasting

Revenue forecasting algorithms were implemented to analyse the revenue generation trends for subscription-based businesses. Revenue forecasting was performed to analyse the trends in revenue generation for the subscription-based business model. Using the customer subscription charges and customer tenure variables, the revenue generation trends were analysed to identify the factors that influence revenue generation for the subscription-based business model.

Monthly Recurring Revenue (MRR) was computed to identify the total predictable revenue generated from the customer base for a given period. The revenue forecasting model will help to identify the sustainability of the revenue generation model for the subscription-based business model.

4. Unit Economics Analysis

Analysis of unit economics was performed in the study with the objective of determining the financial viability of customer acquisition and retention strategies. Unit economics analysis is based on the most critical SaaS financial metrics, including Customer Lifetime Value (LTV), Customer Acquisition Cost (CAC), Contribution Margin (CM), and the LTV to CAC ratio. These metrics are critical in the analysis of the financial viability of the revenue generated from customers in comparison with the cost of acquisition and delivery of the product or service. The analysis of the unit economics of the SaaS business provides critical insights into the financial performance of the SaaS business model.

5. Integration of Analytical Techniques

The integration of descriptive analysis, predictive modelling, revenue forecasting, and unit economics analysis provides a critical analysis of the SaaS business model. The integration of behavioural analysis and unit economics analysis provides critical insights into the application of predictive analytics and unit economics optimization in the development of sustainable SaaS growth strategies.

Chapter 5 – Data Analysis and Interpretation

5.1 Descriptive Analysis

The descriptive analysis was carried out to identify the underlying patterns of the customer subscription behaviour and churn aspects within the data. This analysis has been carried out using the Python programming language in the Google Collab environment. Pandas and NumPy libraries were used for data manipulation, while data visualization has been carried out using the Matplotlib and Seaborn libraries.

Descriptive statistical methods and data visualization techniques have been employed to understand the data regarding aspects such as churn, tenure, pricing, and contract types for the subscribers. This helps to understand the underlying factors of the customer behaviour patterns, which may lead to churn, before moving to the predictive modelling analysis.

1. Distribution of Customer Churn

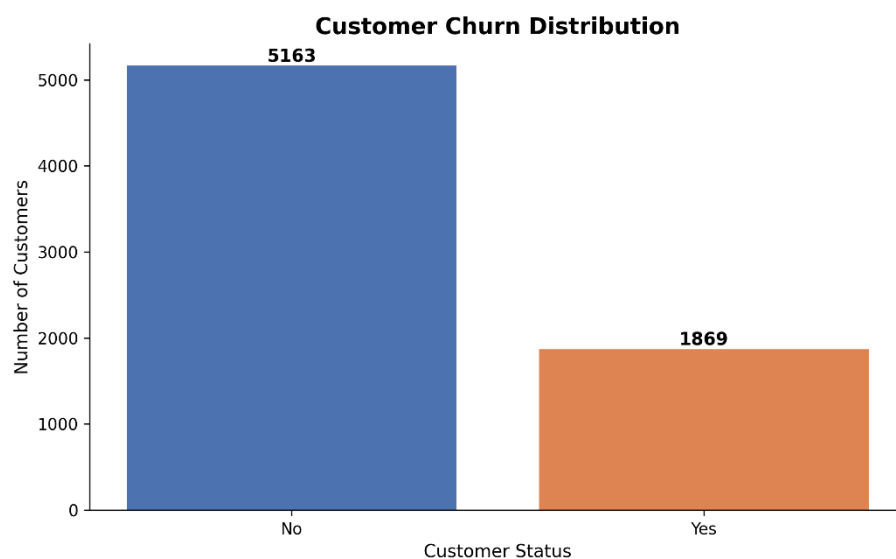


Figure 5.1: Customer Churn Distribution

Figure 5.1 presents the distribution of customers based on their churn status. The results show that out of the total customer base, 5,163 customers did not churn, while 1,869 customers discontinued the service. This indicates that approximately 73% of customers remained subscribed, whereas about 27% of customers churned. The dataset therefore exhibits a moderate level of churn, which is typical for subscription-based service industries and provides a suitable basis for predictive modelling analysis.

2. Contract Type and Customer Churn

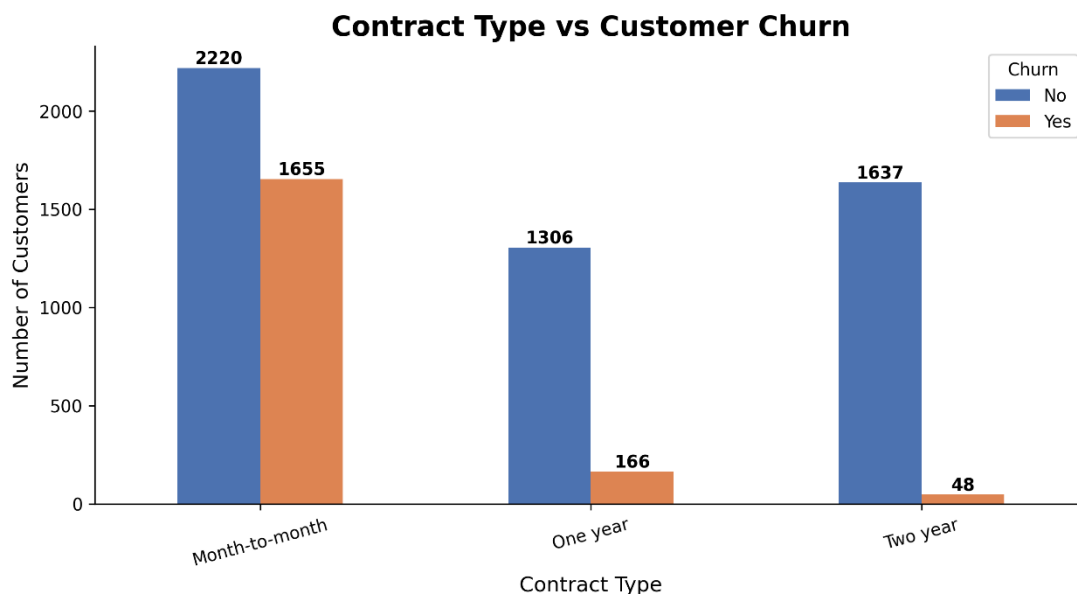


Figure 5.2: Contract Type vs Customer Churn

The relationship between contract type and customer churn is illustrated in Figure 5.2. The results reveal that customers with month-to-month contracts show the highest churn count, with 1,655 customers leaving the service, compared to 2,220 customers who remained. In contrast, customers with longer-term contracts demonstrate significantly lower churn levels. Among customers with one-year contracts, only 166 customers churned compared to 1,306 retained customers. Similarly, among customers with two-year contracts, churn was extremely low with only 48 customers leaving the service compared to 1,637 retained customers. This pattern suggests that longer contractual commitments are associated with higher customer retention.

3. Churn Percentage by Contract Type

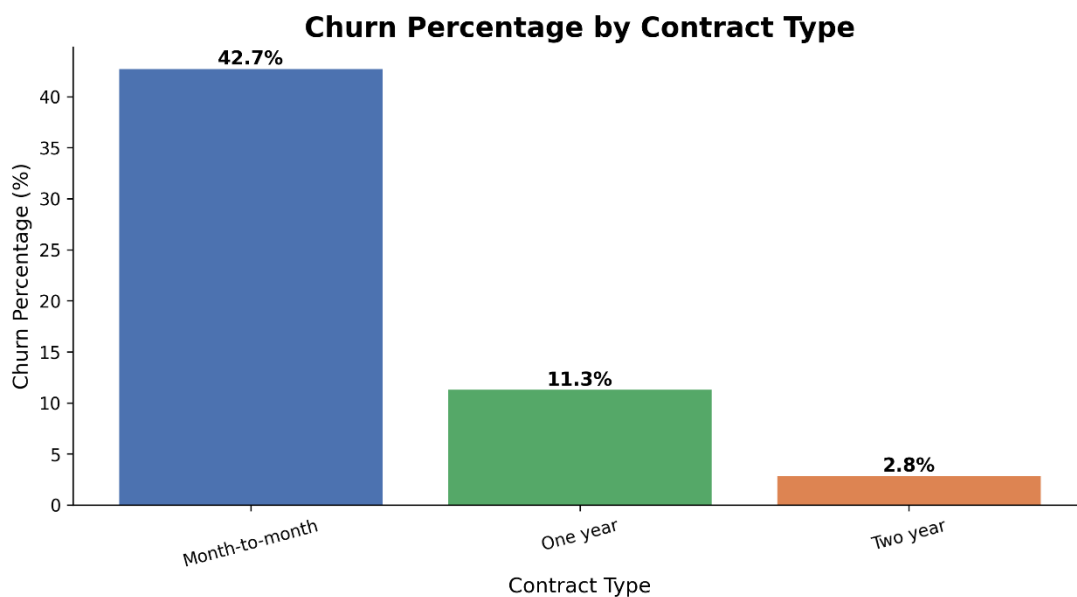


Figure 5.3: Churn Percentage by Contract Type

To further understand the relative churn risk across contract types, Figure 5.3 presents churn percentages for each contract category. The analysis indicates that approximately 42.7% of month-to-month customers churned, which represents the highest churn rate among all contract types. In comparison, the churn rate among one-year contract customers was approximately 11.3%, while the churn rate for two-year contract customers was only about 2.8%. These findings indicate that longer contract durations significantly reduce the likelihood of customer churn, highlighting the importance of customer commitment mechanisms in subscription-based business models.

4. Customer Tenure Distribution

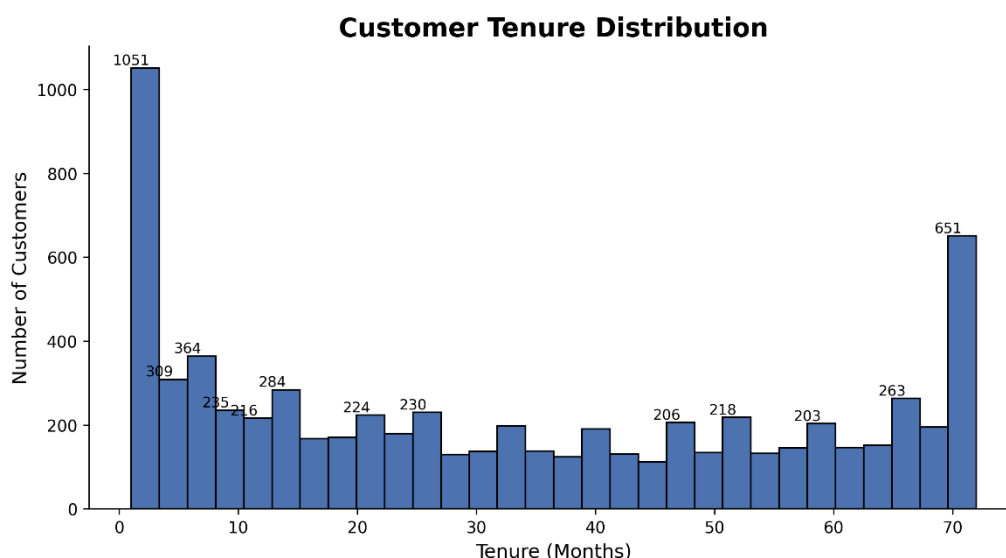


Figure 5.4: Customer Tenure Distribution

Customer tenure distribution is illustrated in Figure 5.4. The tenure variable represents the number of months a customer has remained subscribed to the service. The distribution indicates that a substantial proportion of customers have relatively short subscription durations, while another segment of customers demonstrates long-term engagement with the service. This distribution suggests the presence of both newly acquired customers and long-standing customers within the dataset, which is important for understanding lifecycle patterns in subscription-based businesses.

5. Customer Tenure and Churn Behaviour

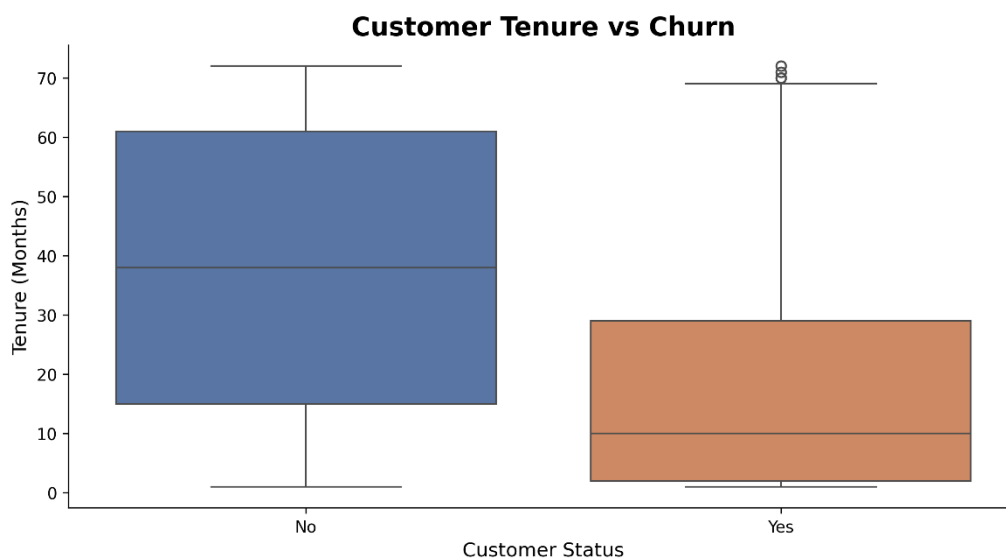


Figure 5.5: Customer Tenure vs Customer Churn

Figure 5.5 examines the relationship between customer tenure and churn behaviour. The boxplot clearly indicates that customers who churn tend to have significantly shorter tenure compared to customers who remain subscribed. The median tenure of retained customers appears substantially higher than that of churned customers, indicating that customers are more likely to leave the service during the early stages of their subscription lifecycle. In contrast, customers with longer tenure tend to exhibit greater loyalty and are less likely to churn.

6. Monthly Charges Distribution

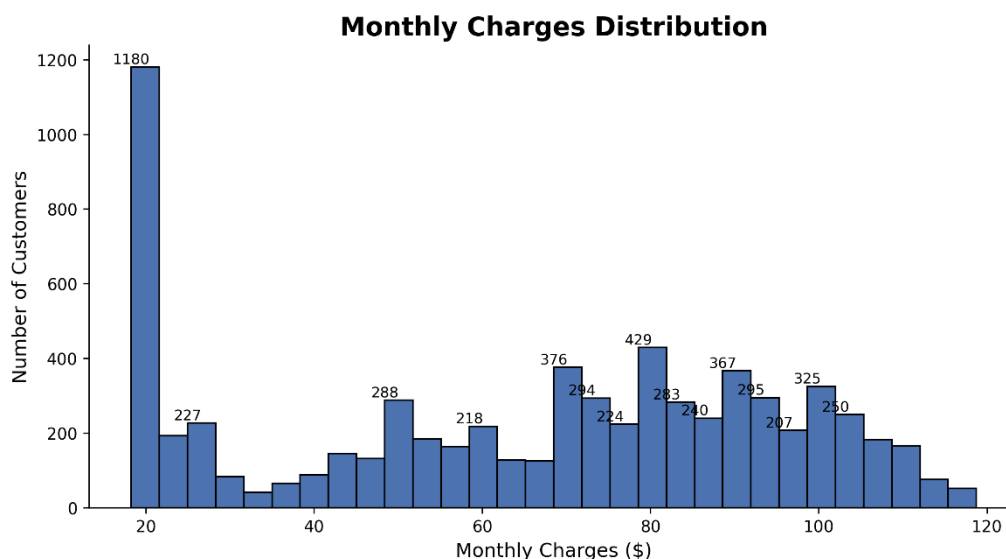


Figure 5.6: Monthly Charges Distribution

Figure 5.6 illustrates the distribution of monthly subscription charges across customers. The histogram shows that monthly charges range approximately from \$18 to \$118. A large proportion of customers appear to fall within the higher pricing ranges, particularly between approximately \$70 and \$100 per month, while another group of customers is concentrated around lower price levels near \$20. This

indicates that the service offers multiple pricing tiers corresponding to different service bundles or product offerings.

7. Monthly Charges and Customer Churn

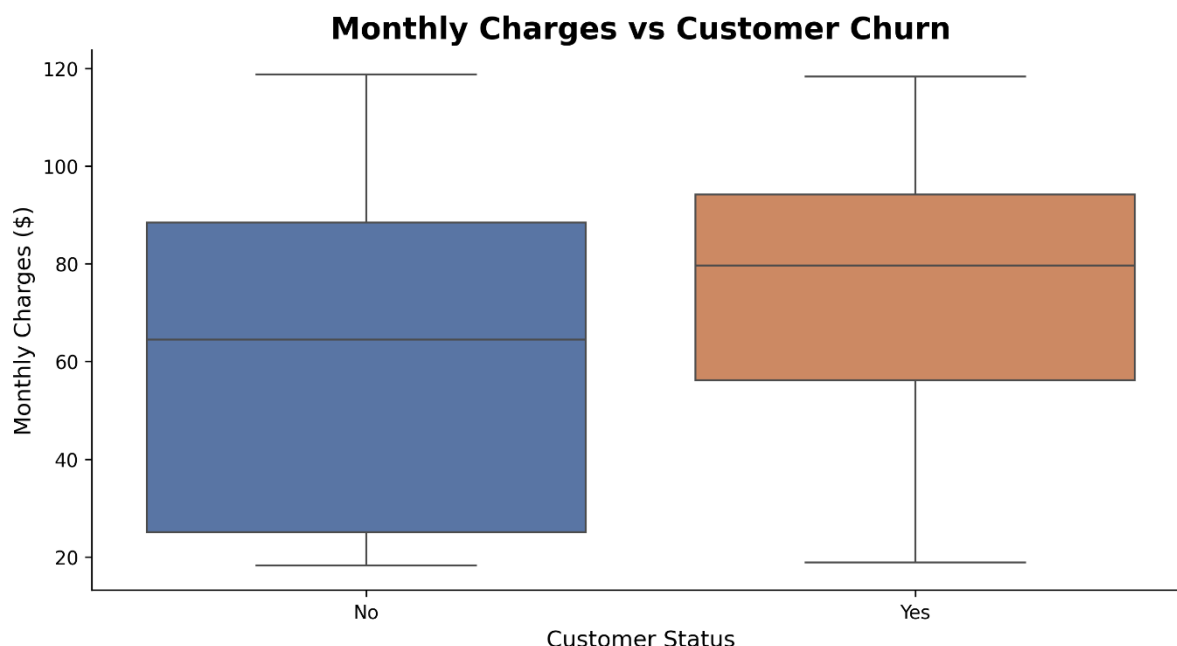


Figure 5.7: Monthly Charges vs Customer Churn

The relationship between monthly subscription charges and churn behaviour is shown in Figure 5.7. The boxplot reveals that customers who churn tend to have higher median monthly charges compared to customers who remain subscribed. This suggests that higher pricing levels may contribute to increased churn risk, possibly due to greater price sensitivity among customers or perceived differences in value received from the service. However, the overlap between the distributions indicates that pricing alone does not fully determine churn behaviour, and other factors such as contract type and customer tenure also play an important role in influencing customer retention.

5.2 Churn Prediction Model

This section will include the results obtained from the predictive modelling analysis, which aims to determine the probability of customer churn using machine learning models. This analysis will be used to assess the performance of the predictive models in identifying the likelihood of customer churn. Two supervised learning models were developed and implemented to analyse the likelihood of customer churn: the Logistic Regression model and the Random Forest model.

The models were trained on the dataset, considering the variables such as customer tenure, monthly charges, contract type, payment method, and internet service type. The dataset was divided into training and testing sets, with an 80:20 ratio, to assess the performance of the models on the testing data set. The target variable considered for the model was the churn status of the customer.

The performance of the models developed was assessed based on the classification accuracy and the results obtained from the analysis of the confusion matrix. The results obtained from the analysis will be compared with the actual churn status of the customer.

1. Logistic Regression Model

Logistic Regression was first used as a baseline model to classify the data and make predictions regarding customer churn. It is a statistical method commonly used to solve binary classification problems where there are only two possibilities or outcomes for a dependent variable. In this case, it is either churn or no churn.

Logistic Regression results showed an accuracy of approximately 78.6% when it came to classifying the data. This means that out of every five cases of customers, it correctly classified almost four of them.

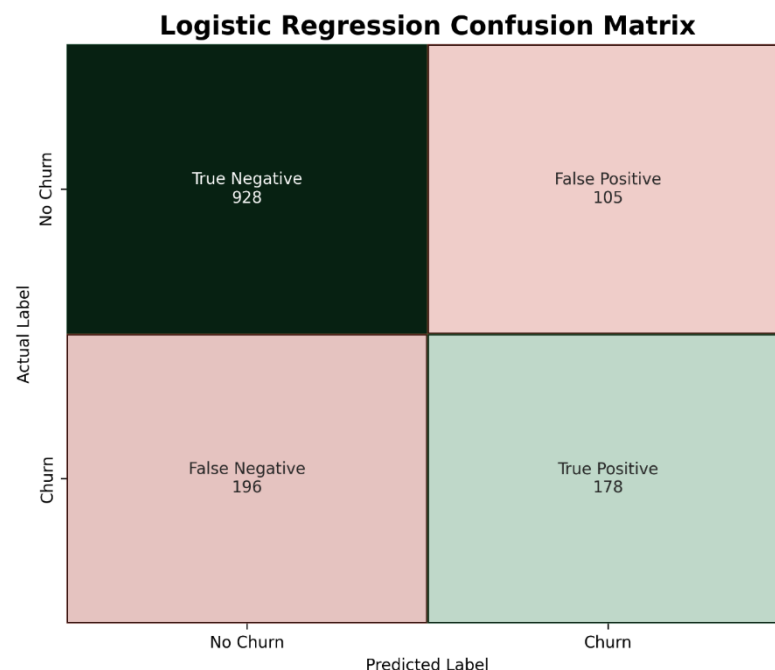


Figure 5.8: Logistic Regression Confusion Matrix

Figure 5.8 illustrates a confusion matrix of the Logistic Regression model. From this confusion matrix, it is evident that it correctly classified 928 customers who did not churn and also successfully predicted 178 customers who actually churned. However, it incorrectly classified 105 customers who were still subscribed to the company's services when they were supposed to churn. It also failed to predict 196 customers who ended up churning.

It is evident from the results that the Logistic Regression model is able to classify non-churn customers quite well but has moderate limitations when it comes to predicting all cases of churners. This is a common limitation when dealing with datasets that have more cases of non-churners compared to churners.

2. Random Forest Model

In addition to Logistic Regression, another classification model of Random Forest was implemented to address potential nonlinear relationships between customer attributes and churn behaviour. Random Forest is another ensemble learning technique where multiple decision trees are created and combined for better model performance and robustness.

The performance of the Random Forest model was measured in terms of overall accuracy and achieved approximately 76.1%. Although slightly lower in terms of overall accuracy compared to the Logistic Regression model, the performance of the Random Forest model was competitive in terms of overall accuracy.

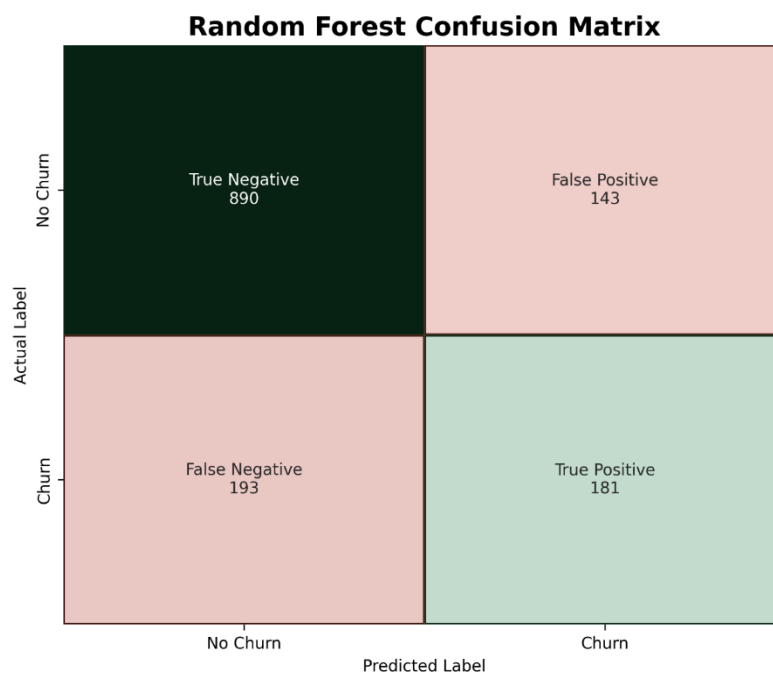


Figure 5.9: Random Forest Confusion Matrix

The confusion matrix for the Random Forest model is shown in Figure 5.9. From the output of the model, it is observed that the model was able to identify 890 customers who did not churn and was successful in identifying 181 customers who churned. On the other hand, it was observed that the model incorrectly predicted 143 customers as churners who actually did not churn and was not able to identify 193 customers who actually churned in the future.

From the above two models, it is observed that although the Logistic Regression model performed slightly better in terms of overall accuracy, the Random Forest model was able to identify slightly more churners compared to Logistic Regression. In actual scenarios of churn management, identifying potential churners is more valuable for business organizations.

3. Feature Importance Analysis

To gain more insight into the major drivers of churn prediction, feature importance analysis was carried out using the Random Forest model. Feature importance is a measure of the importance of each input feature in predicting customer churn.

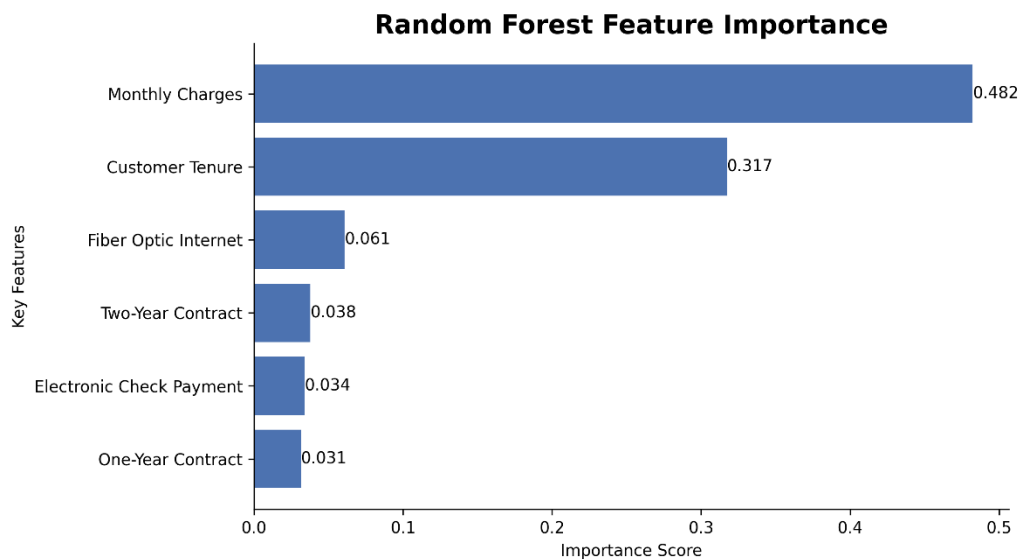


Figure 5.10: Random Forest Feature Importance

As shown in Figure 5.10, it can be observed that Monthly Charges and Customer Tenure are the major drivers of customer churn prediction using the Random Forest model.

According to the results, Monthly Charges have the highest importance score of about 0.482, which shows that pricing levels are major drivers of customer churn prediction. Customer Tenure is the second major driver of customer churn prediction, as it has an importance score of about 0.317. This shows that customers with shorter tenure are more likely to churn.

The next major drivers of customer churn prediction are Fiber Optic Internet Service, which has an importance score of about 0.061; Two-Year Contract, which has an importance score of about 0.038; Electronic Check Payment, which has an importance score of about 0.034; and One-Year Contract, which has an importance score of about 0.031.

This is similar to the patterns observed during the descriptive analysis. The patterns showed that customers with higher monthly charges and those with shorter tenure are more likely to churn, while those with longer contractual commitments are less likely to churn.

5.3 Revenue Forecasting

This section of the report discusses the revenue forecasting analysis carried out to determine the revenue potential of the subscription-based business model using the Monthly Recurring Revenue (MRR) concept. Monthly Recurring Revenue is an essential financial metric for Software-as-a-Service (SaaS) subscription-based business models, which represent the revenue earned from the customer base.

Revenue forecasting for Software-as-a-Service business models is carried out by determining the present recurring revenue and projecting future revenue based on growth and retention rates. Revenue

projection for this study is determined based on the present revenue earned from the subscription-based business model for the customer base and the growth in recurring revenue over time.

The revenue forecasting analysis for this study has been carried out in three steps. First, the present Monthly Recurring Revenue (MRR) earned from the customer base is determined. Then, the financial effect of customer churn is analysed. Finally, revenue projection for the next five years is determined based on the growth in revenue.

1. Monthly Recurring Revenue (MRR)

Monthly Recurring Revenue is defined as the total revenue earned from all the active customers during a given month. It is calculated by summing all the monthly subscription amounts from all the customers. Formula:

$$MRR_t = \sum_{i=1}^n Monthly_Revenue_i$$

According to the analysis above, it is evident that the total Monthly Recurring Revenue earned from all the existing customers is approximately \$455,661 each month. This figure represents the revenue earned from the subscription-based service before any growth or churn effects are considered.



Figure 5.11: Monthly Recurring Revenue

The calculated Monthly Recurring Revenue represents the revenue earned from the subscription-based service. It is an essential factor that is considered when forecasting revenue. SaaS-based organizations rely on the revenue earned from subscriptions. Hence, Monthly Recurring Revenue is considered an essential factor when evaluating revenue growth potential.

Having a stable revenue stream is essential when forecasting revenue growth. SaaS-based organizations can easily forecast revenue growth compared to other organizations with transaction-based revenue models. Hence, the calculated Monthly Recurring Revenue is an essential factor when evaluating the financial viability of the organization.

2. Revenue Impact of Customer Churn

Customer churn has a direct impact on revenue stability. If customers leave a subscription-based service, it means that revenue from that customer is no longer available to the company. Hence, it is important to understand the financial impact of customer churn to determine the viability of revenue models of SaaS companies.

In this analysis, revenue is divided into revenue from retained customers who have not churned and revenue from customers who have churned. From the results of this analysis, it is evident that \$316,530 of monthly revenue is from retained customers who have not churned, while \$139,131 of monthly revenue is from customers who have churned.

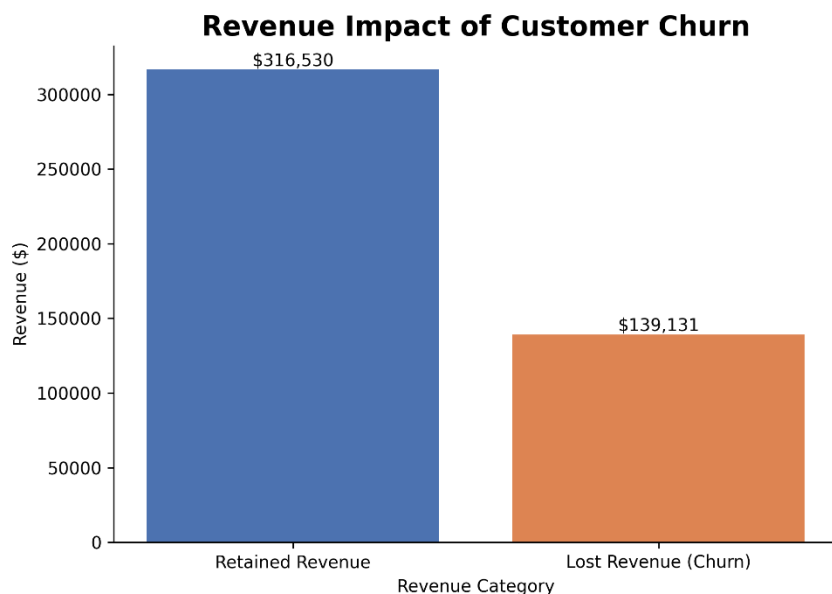


Figure 5.12: Revenue Impact of Customer Churn

These results indicate that a large portion of revenue is not being utilized by the company due to customer churn. From the results of this analysis, it is evident that 30.5% of Monthly Recurring Revenue is from revenue associated with customers who have churned, while 69.5% of Monthly Recurring Revenue is from retained customers who have not churned.

These results indicate that customer retention is very important to the financial success of SaaS companies. If customer churn is reduced, revenue would increase without any additional cost of acquiring new customers. Hence, customer retention would have a direct impact on the financial success of SaaS companies.

3. Projected Revenue Growth

To determine the possible growth in subscription revenue, a five-year revenue projection model was created using the current level of Monthly Recurring Revenue and a projected annual growth rate of 20 percent. This growth rate is a reasonable assumption of business growth potential that is typically experienced in a subscription-based business model.

A compound growth projection model is utilized to determine future levels of revenue that can be generated in a five-year projection. Based on this projection, it is possible to determine that future levels of subscription revenue can grow from a current level of monthly revenue to a projected level of \$546,793 in Year 1. As a result of continued growth in this business model, projected future levels

of revenue are expected to increase to a level of \$656,152 (Year 2), \$787,382 (Year 3), \$944,859 (Year 4), and finally to a level of \$1,133,830 (Year 5).

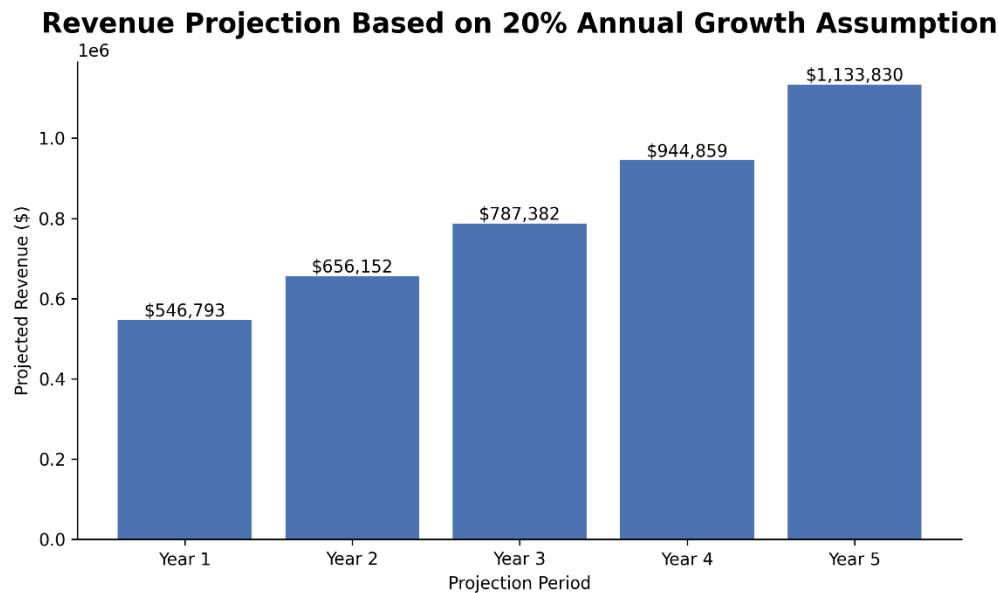


Figure 5.13: Projected Revenue Growth

The projected trend indicates the compounded growth of the revenue generated over time. This indicates the advantages of the SaaS revenue model, which enables the business to grow its revenue more efficiently than traditional models. This analysis of the revenue forecasting model indicates the advantages of having strong revenue and reducing the churn rate, which can positively impact the sustainability of the business.

The analysis of the revenue forecasting model indicates that having strong revenue and minimizing the churn rate can positively impact the sustainability of the subscription-based business model. This analysis supports the need for churn prediction and retaining the existing customer base, which was discussed in the previous section. Reducing the churn rate can positively impact the future revenue potential of the business.

5.4 Unit Economics Analysis

This section seeks to analyse the financial sustainability of the subscription business model using unit economics analysis. Unit economics is defined as the financial value generated by each individual customer compared to the cost incurred to acquire the customer. In the context of the SaaS model, unit economics is often measured using Customer Lifetime Value, Customer Acquisition Cost, and the LTV/CAC Ratio.

Customer Lifetime Value is the revenue earned from customers throughout their subscription tenure. This metric offers an idea of the amount of business generated from each customer. Customer Acquisition Cost is the average amount of expenses incurred by the company to acquire new customers through marketing and sales initiatives. The interrelation between LTV and CAC is often assessed using LTV/CAC.

In this research, Customer Lifetime Value was determined based on the amount of subscription charges incurred by the customer and their subscription tenure.

1. Customer Lifetime Value Calculation

The LTV formula used in the analysis:

$$LTV = Monthly_Revenue \times Tenure$$

Using this method, the Customer Lifetime Value for all the customers in the dataset was determined by multiplying the monthly subscription cost by the number of months for which the customer stayed subscribed to the service. The distribution of Customer Lifetime Value for all the customers in the dataset is represented in Figure 5.14.

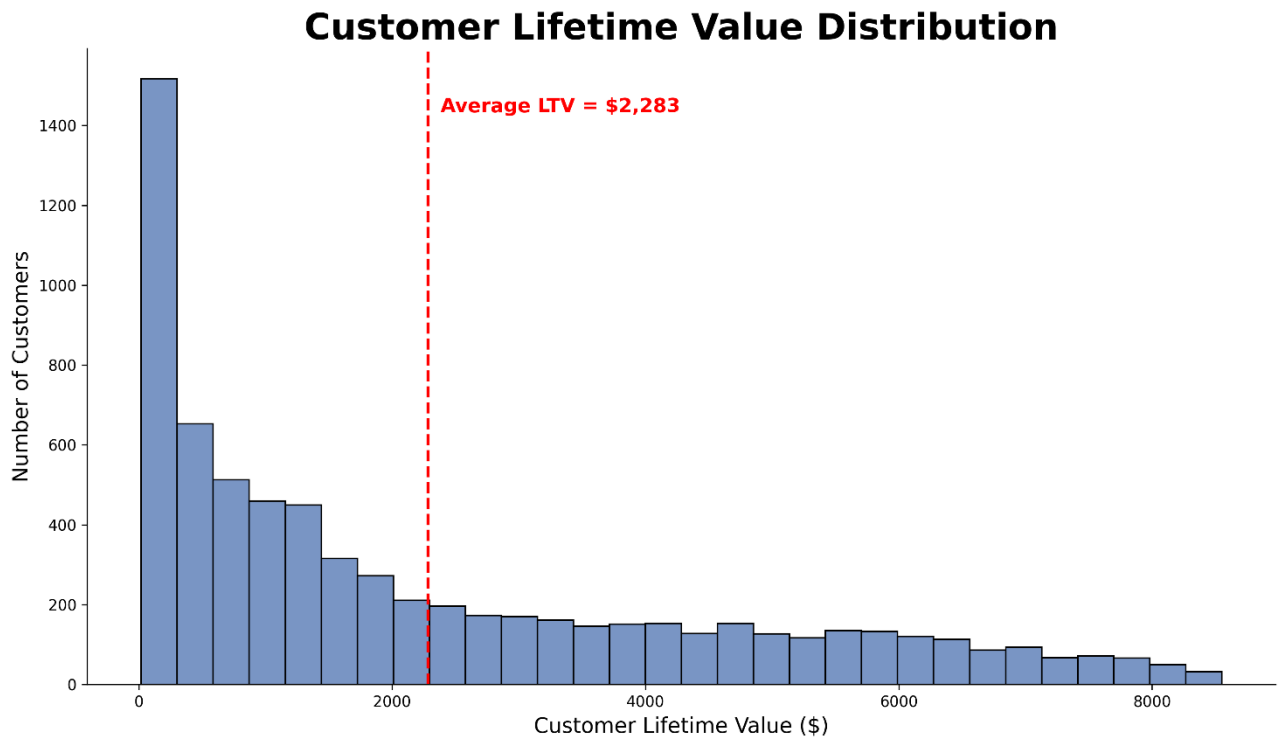


Figure 5.14: Customer Lifetime Value Distribution

The distribution of Customer Lifetime Value for all the customers in the dataset shows a high level of variability in terms of customer value. While some customers have a low Customer Lifetime Value due to a low subscription period, others have a high Customer Lifetime Value since they stayed subscribed for a longer period of time. The analysis of the data reveals that the average Customer Lifetime Value for all the customers in the dataset is around \$2,283.

2. Customer Acquisition Cost (CAC)

Customer Acquisition Cost is defined as the average cost of acquiring a new customer through marketing and sales efforts. Since marketing expenditure data is not available in the data set considered for this research, CAC is estimated using a reasonable hypothesis based on industry norms.

For this research, it is considered that the Customer Acquisition Cost is \$350 per customer. This is an approximate average cost of acquiring a new customer, which is considered typical in the digital service-based business model.

$$CAC_i = k \times Monthly_Revenue_i$$

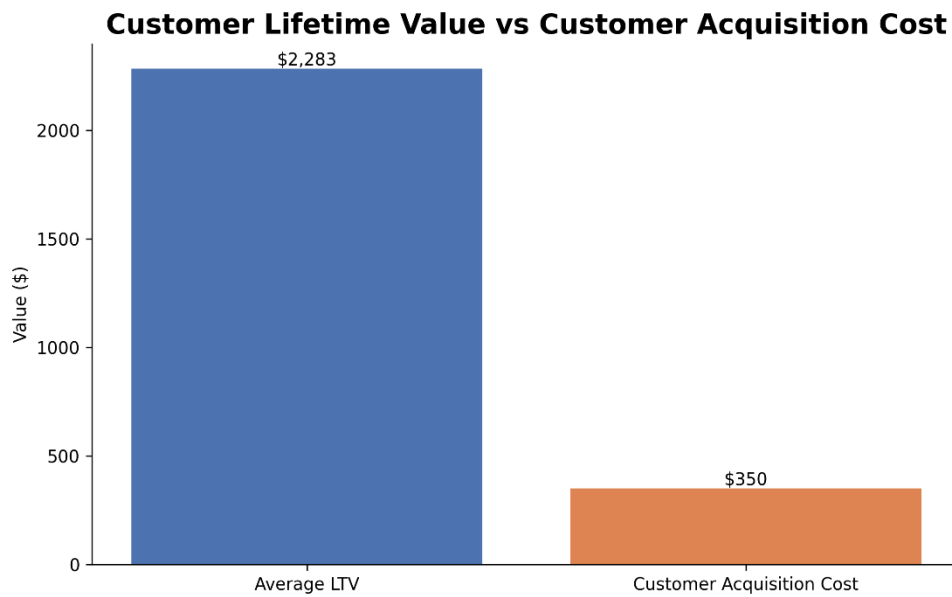


Figure 5.15: Customer Lifetime Value vs Customer Acquisition Cost

To validate the financial viability of this business model, the average Customer Lifetime Value is compared with the Customer Acquisition Cost. The comparison between LTV and CAC is shown in Figure 5.15.

Based on this comparison, it is observed that the average Customer Lifetime Value is \$2,283, which is significantly higher than the Customer Acquisition Cost of \$350. This shows that the revenue earned from customers during their lifetime is significantly higher than the amount spent on acquiring new customers.

3. Contribution Margin Analysis

Contribution margin is the measure of revenue available after deducting service delivery costs. Contribution margin is an essential metric to measure revenue generated from each customer to meet fixed expenses and generate profit. Contribution margin is an essential metric to measure the efficiency of a SaaS-based subscription service.

Contribution margin is estimated on the basis of average monthly revenue generated from each customer compared to service delivery costs. From the dataset provided, average monthly revenue generated from each customer is approximately \$64.80, referred to as Average Revenue Per User (ARPU). Since no direct information is available on infrastructure and service delivery costs from the given dataset, it is considered that 40% of monthly revenue is service delivery cost, as it is a common practice for SaaS-based digital services. Using this assumption, it is estimated that \$25.92 is service delivery cost, and \$38.88 is contribution margin per customer.

$$CM_i = \text{Monthly_Revenue}_i - \text{Service_Cost}_i$$

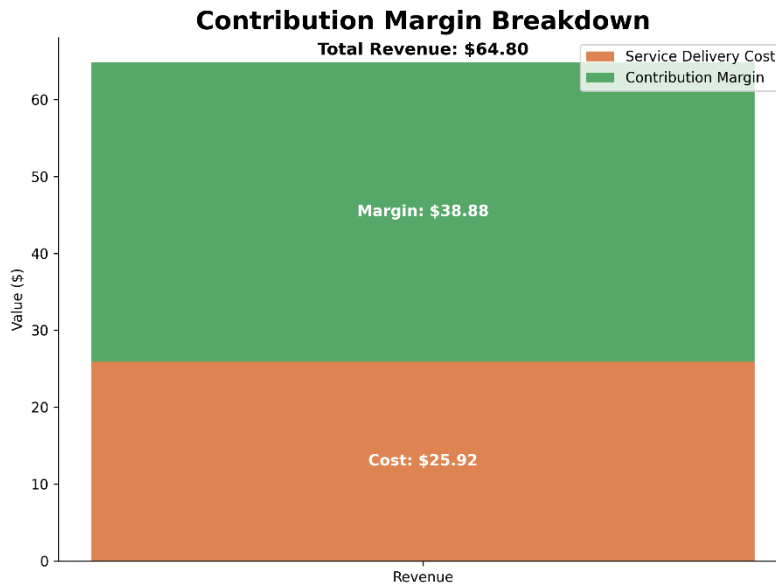


Figure 5.16: Contribution Margin Breakdown

Figure 5.17: Breakdown of average monthly revenue into service delivery cost and contribution margin. As per the results provided above, it is evident that 60% of monthly revenue is available as contribution margin, indicating that a major portion of revenue generated from each customer is available to meet customer acquisition costs and generate profit.

Contribution margin of 60% indicates good operational efficiency of a SaaS-based subscription service. It is considered that SaaS-based digital services with 50-70% contribution margin are considered to be financially viable. It is evident from the results provided above that the subscription service has good potential to generate profit from each customer relationship provided that customer acquisition costs remain under control.

4. LTV: CAC Ratio Analysis

The efficiency of SaaS unit economics is typically evaluated using the LTV:CAC ratio:

$$LTV : CAC = \frac{LTV}{CAC}$$

Using the values calculated above, it is noted that in this analysis, the LTV:CAC ratio is approximately 6.52. This means that for every dollar spent in acquiring a customer, the company is able to generate approximately \$6.52 in lifetime revenue for that particular customer.

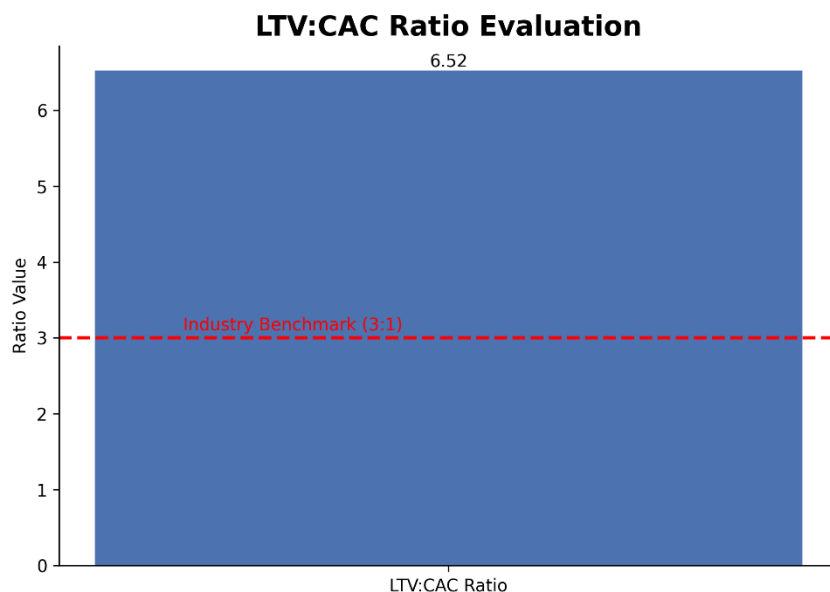


Figure 5.17: LTV Ratio Evaluation

In SaaS industry benchmarks, it is noted that for SaaS companies, it is ideal to have an LTV:CAC ratio of approximately 3:1 for healthy and sustainable business growth in the future. Any ratios less than that may represent inefficient spending in acquiring customers for the business, while ratios significantly higher may represent healthy and efficient spending in acquiring customers.

The calculated LTV:CAC ratio of 6.52 in this study shows that it is highly likely that the business model is demonstrating healthy unit economics since it is able to generate significantly more in lifetime revenue compared to the cost of acquiring customers.

Financial Sustainability Insights

The evaluation of unit economics helps to gain critical insights into the financial sustainability of the subscription-based business model under investigation in this study. The critical findings obtained from the analysis are provided below:

- High Customer Lifetime Value

The estimated Customer Lifetime Value is found to be \$2,283 on average. This indicates that the customers generate a high amount of revenue for the business throughout their subscription period. The longer the customer stays with the business, the higher amount of revenue is obtained from them compared to those who leave the business at an early stage of their subscription period.

- Efficient Customer Acquisition Cost

The estimated Customer Acquisition Cost is found to be \$350 per customer. This is a very low amount compared to the overall revenue obtained from each customer throughout their subscription period with the business.

- Healthy Contribution Margin (CM)

The analysis of contribution margin reveals that the monthly revenue obtained from each customer is \$64.80 on an average, whereas the cost-of-service delivery to each customer is \$25.92. This results in

a contribution margin of \$38.88 per customer per month, which is around 60 percent of the contribution margin ratio.

- Strong LTV Ratio

The LTV ratio, which has been determined, stands at approximately 6.52, implying that the company makes approximately 6.52 dollars for each dollar spent on acquiring the customer. Based on industry analysis, the LTV ratio of 3:1 for SaaS companies is considered good. This, therefore, suggests that the LTV ratio for the company under analysis is strong.

- Operational Efficiency of the Subscription Model

A contribution margin of approximately 60% suggests that the subscription service has the ability to keep a significant portion of the generated revenue even after covering the variable costs. SaaS companies with contribution margins ranging from 50% to 70% are considered operationally efficient, and the business model under analysis appears to be operationally efficient, suggesting good scalability potential.

- Impact of Customer Retention on Financial Performance

The results obtained from the churn prediction models indicated that customer tenure and monthly charges are the main factors affecting churn. Since customer tenure affects the LTV ratio, improving retention strategies would be beneficial for the company.

- Integration of Predictive Analytics and Unit Economics

The integration of the results obtained from the churn prediction models and the unit economics metrics suggests how the customer affects the financial performance of the company. Predictive models can be used to identify high-risk customers, and the unit economics metrics will be used to evaluate the financial impact of retaining or losing such high-risk customers.

Chapter 6 – Major Findings and Discussion

6.1 Overview of the Study

The purpose of the present study was to assess the financial sustainability of the subscription-based SaaS business model using the combination of predictive analytics and unit economics analysis. To achieve the purpose of the study, the researchers conducted descriptive analysis, churn prediction modelling, revenue forecasting, and unit economics analysis using the given dataset on customer subscriptions.

The findings of the study offer an insight into the use of predictive modelling techniques in conjunction with financial metrics such as Customer Lifetime Value (LTV), Customer Acquisition Cost (CAC), and Contribution Margin (CM).

6.2 Major Findings of the Study

The major findings obtained from the data analysis are presented in the following sections.

1. Customer Churn Patterns

- Contract types have a significant impact on customer churn patterns.

Customers with a month-to-month contract showed the highest customer churn rate at 42.7%, while

customers with a one-year contract and two-year contract showed significantly low customer churn rates at 11.3% and 2.8%, respectively.

- Customer tenure is also highly related to customer churn probability.

Customers with low subscription periods showed high customer churn, while customers with long subscription periods showed high customer loyalty.

- Pricing also has a significant impact on customer retention.

Customers with high monthly charges showed a slightly high customer churn probability, and hence pricing may be a factor in customer satisfaction.

2. Predictive Modelling Performance

Two different models were developed using a machine learning model to predict customer churn.

Logistic Regression Model

Random Forest Model

- The model developed using the logistic regression method showed an accuracy level of 78.6%, and hence the model is effective in predicting potential customer churn.
- The model developed using the random forest method also showed significant results of 76.1%, and feature importance analysis was also performed.
- From the feature importance analysis, it is clear that the monthly charges and customer tenure are highly related to customer churn.

3. Revenue Forecasting Insights

- The Monthly Recurring Revenue (MRR) is approximately \$455,661.
- Based on projected growth assumptions, the revenue forecast suggests that total revenue could increase from:

Year	Projected Revenue
Year 1	\$546,793
Year 2	\$656,152
Year 3	\$787,382
Year 4	\$944,859
Year 5	\$1,133,830

- The revenue growth is consistent over time, indicating that the business model is scalable to achieve more financial growth in the future.
- Revenue forecasting analysis revealed that customer churn results in a loss of \$139,131 in revenue, indicating that customer retention is critical to business success.

4. Unit Economics and Financial Sustainability

- The Customer Lifetime Value (LTV) is approximately \$2,283, indicating that the business model generates considerable revenue from individual customers over time.
- The Customer Acquisition Cost (CAC) is approximately \$350, indicating that acquiring customers is not costly for the business model, given the considerable revenue generated from individual customers over time.
- The LTV: CAC is approximately 6.52, indicating that every dollar spent on acquiring customers translates to \$6.52 in customer lifetime value.
- Contribution margin analysis indicated:

Metric	Value
Average Monthly Revenue	\$64.80
Service Delivery Cost	\$25.92
Contribution Margin	\$38.88
Contribution Margin Ratio	≈ 60%

- The business model has a high contribution margin of approximately 60%, indicating that the business model is efficient in its operations because a significant portion of revenue is left after paying for service delivery costs.

6.3 Discussion of Findings

The results of this research emphasize the importance of considering predictive analytics results along with financial metrics while assessing a SaaS business model. In most financial models, only growth rate is considered; however, for a subscription-based business, other factors must be analysed as well.

As observed from the results of the customer churn prediction analysis, customer tenure and contract type are major factors affecting customer retention. These results are consistent with the notion that a SaaS business should strive to retain customers for a longer period of time.

The results of the revenue forecasting analysis revealed that customer churn has a major financial impact on a business. These results indicate that predictive modelling can be a powerful tool for a business to predict customer churn and take necessary steps to retain customers.

The results of unit economics analysis are indicative of a financially viable business model. The LTV to CAC ratio is very high, which indicates that a business is able to leverage a lot of value from each new customer acquired. In addition, a healthy contribution margin indicates that a business is able to grow operations while keeping costs low.

6.4 Implications for SaaS Business Strategy

The implications of the research study for business strategy are as follows:

- Customer retention should be given prime importance because this can result in higher revenues and profitability for the business.
- Long-term subscription contracts should be encouraged to reduce customer churn rates significantly.

- Predictive analytics can be used to identify customer behaviour that may lead to customer churn.
- The LTV: CAC should be maintained to ensure financial sustainability for the business in the long term.
- Improving contribution margins through optimization can strengthen profitability.

Chapter 7- Conclusion and Recommendations

7.1 Conclusion

The objective of the present study is to evaluate the financial sustainability of the SaaS business model by integrating the concept of predictive analytics with unit economics evaluation. To achieve the objective of the study, the researchers employed a dataset representing the customer behaviour of the SaaS business model. The study analysed the relationship between customer behaviour and the long-term financial performance of the SaaS business model through descriptive analysis, prediction modelling, revenue forecasting, and unit economics evaluation.

The descriptive analysis of the study identified a number of interesting facts about the customer behaviour. The study observed a strong relationship between the type of contract and customer churn. The study revealed that customers with a month-to-month subscription contract have a significantly higher probability of churn compared to customers with a long-term subscription contract. The study observed a strong relationship between customer tenure and churn. It was revealed that customers with a shorter tenure period are more likely to churn from the service. This implies that the loyalty of customers increases over time.

The prediction modelling performed by the study revealed that the SaaS business model can employ the concept of predictive analytics to predict the customers who are likely to churn from the service. The study employed the logistic regression algorithm to develop a prediction model. The study revealed that the logistic regression model has a reasonable accuracy of about 78.6%. The study further revealed that the monthly charges and customer tenure are the most influential factors that affect the customer churn.

The results of the revenue forecasting also revealed the scalability potential of a subscription-based business model. The results of the forecasting revealed that the Monthly Recurring Revenue (MRR) is approximately \$455,661. The five-year revenue forecasting results revealed that a subscription-based business has a great potential for long-term financial growth if customer retention is well managed.

The unit economics results provided additional insights into the financial sustainability of a subscription-based business model. The results revealed that the average Customer Lifetime Value (LTV) is approximately \$2,283. The results also revealed that the Customer Acquisition Cost (CAC) is \$350. The results revealed that the LTV to CAC ratio is approximately 6.52. The results also revealed that the contribution margin is approximately 60%. The results revealed that a business is able to retain a significant amount of income after meeting the cost of delivering services to customers.

The findings of this study revealed that a predictive modelling approach is important in creating a framework for evaluating the financial sustainability of a subscription-based business model. The results revealed that a business can use unit economics metrics to inform strategic decisions that are aimed at improving customer retention and financial growth.

7.2 Managerial Recommendations

The findings of this research provide various recommendations for SaaS firms that want to enhance their financial sustainability.

1. Customer Retention Strategies

- Firms must consider customer retention strategies as it has a significant effect on enhancing overall revenue and profitability.
- Firms must provide incentives to customers to shift from month-to-month subscriptions to long-term subscriptions, which can significantly reduce customer churn rate.

2. Predictive Analytics for Churn Management

- Firms must use predictive customer churn management tools that use machine learning algorithms to predict customer churn risk.
- Predicting customer churn risk can help firms use various strategies to retain customers by identifying potential customers who are at a higher risk of churning from their services.

3. Pricing and Revenue Optimization

- Firms must consider pricing strategies that can enhance customer satisfaction as well as maximize their revenue.
- Firms must monitor the relationship between monthly charges and customer churn rate to enhance pricing structures that can maximize customer retention.

4. Improving Unit Economics

- Firms must monitor Customer Lifetime Value (CLV), Customer Acquisition Cost (CAC), and Contribution Margin to enhance financial sustainability.
- Firms must ensure that the LTV/CAC ratio is above 3:1 to ensure that customer acquisition is profitable.
- Firms must improve operational efficiency to enhance profitability.

7.3 Limitations of the Study

However, despite the valuable insights being generated from the research, there are a number of limitations that need to be taken into consideration:

- The data set used in the study is a proxy for SaaS subscription. It is not actual data from a SaaS company. Therefore, the actual costs incurred in delivering the service and the marketing costs had to be approximated on the basis of the general industry benchmarks.
- The predictive models created from the study are based on a limited number of customer data available. There are a number of other metrics that could be taken into consideration to improve the accuracy of the prediction.
- The revenue forecasting analysis created from the study is based on a number of assumptions. However, the actual SaaS businesses may face a number of challenges due to market competition and the overall market conditions.

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